

IIT JAM 2020 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS

60

TOTAL MARKS

100 M

TEST DURATION

180 Min

PAPER PATTERN

MCQ • MSQ • NAT
(60 Ques)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2020 • Q1 • ID: JAM_2020_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 Let $s_n = 1 + \frac{(-1)^n}{n}$, $n \in \mathbb{N}$. Then the sequence $\{s_n\}$ is

- (A) monotonically increasing and is convergent to 1
- (B) monotonically decreasing and is convergent to 1
- (C) neither monotonically increasing nor monotonically decreasing but is convergent to 1
- (D) divergent

Question 2 (JAM 2020 • Q2 • ID: JAM_2020_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $f(x) = 2x^3 - 9x^2 + 7$. Which of the following is true?

- (A) f is one-one in the interval $[-1, 1]$
- (B) f is one-one in the interval $[2, 4]$
- (C) f is NOT one-one in the interval $[-4, 0]$
- (D) f is NOT one-one in the interval $[0, 4]$

Question 3 (JAM 2020 • Q3 • ID: JAM_2020_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Which of the following is FALSE?

(A) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$

(B) $\lim_{x \rightarrow 0^+} \frac{1}{xe^{1/x}} = 0$

(C) $\lim_{x \rightarrow 0^+} \frac{\sin x}{1 + 2x} = 0$

(D) $\lim_{x \rightarrow 0^+} \frac{\cos x}{1 + 2x} = 0$

Question 4 (JAM 2020 • Q4 • ID: JAM_2020_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $f(x, y) = g(y) + xg'(y)$, then

(A) $\frac{\partial f}{\partial x} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(B) $\frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

(C) $\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(D) $\frac{\partial f}{\partial y} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

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Question 5 (JAM 2020 • Q5 • ID: JAM_2020_Q5)

MCQ

+1 M

Neg: -0.33

JAM 2020

MATHEMATICS - MA

Q. 5 If the equation of the tangent plane to the surface $z = 16 - x^2 - y^2$ at the point $P(1, 3, 6)$ is $ax + by + cz + d = 0$, then the value of $|d|$ is

(A) 16

(B) 26

(C) 36

(D) 46

Question 6 (JAM 2020 • Q6 • ID: JAM_2020_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 If the directional derivative of the function $z = y^2 e^{2x}$ at $(2, -1)$ along the unit vector $\vec{b} = \alpha \hat{i} + \beta \hat{j}$ is zero, then $|\alpha + \beta|$ equals

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$

(D) $2\sqrt{2}$

Question 7 (JAM 2020 • Q7 • ID: JAM_2020_Q7)

MCQ

+1 M

Neg: -0.33

Q. 7 If $u = x^3$ and $v = y^2$ transform the differential equation $3x^5 dx - y(y^2 - x^3)dy = 0$ to $\frac{dv}{du} = \frac{\alpha u}{2(u-v)}$, then α is

(A) 4

(B) 2

(C) -2

(D) -4

Question 8 (JAM 2020 • Q8 • ID: JAM_2020_Q8)

MCQ

+1 M

Neg: -0.33

Q. 8 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation given by $T(x, y) = (-x, y)$. Then

(A) $T^{2k} = T$ for all $k \geq 1$ (B) $T^{2k+1} = -T$ for all $k \geq 1$ (C) the range of T^2 is a proper subspace of the range of T (D) the range of T^2 is equal to the range of T

Question 9 (JAM 2020 • Q9 • ID: JAM_2020_Q9)

MCQ

+1 M

Neg: -0.33

Q. 9 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \left(\frac{n+2}{n} \right)^{n^2} x^n$$

is

(A) e^2 (B) $\frac{1}{\sqrt{e}}$ (C) $\frac{1}{e}$ (D) $\frac{1}{e^2}$

Question 10 (JAM 2020 • Q10 • ID: JAM_2020_Q10)

MCQ

+1 M

Neg: -0.33

JAM 2020

MATHEMATICS - MA

Q. 10 Consider the following group under matrix multiplication:

$$H = \left\{ \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} : p, q, r \in \mathbb{R} \right\}.$$

Then the center of the group is isomorphic to

(A) $(\mathbb{R} \setminus \{0\}, \times)$ (B) $(\mathbb{R}, +)$ (C) $(\mathbb{R}^2, +)$ (D) $(\mathbb{R}, +) \times (\mathbb{R} \setminus \{0\}, \times)$ **Q. 11 – Q. 30 carry two marks each.**

Question 11 (JAM 2020 • Q11 • ID: JAM_2020_Q11)

MCQ

+2 M

Neg: -0.67

Q. 11 Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $l = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$. Which of the following is true?(A) If $l = 1$, then $\lim_{n \rightarrow \infty} a_n = 1$ (B) If $l = 1$, then $\lim_{n \rightarrow \infty} a_n = 0$ (C) If $l < 1$, then $\lim_{n \rightarrow \infty} a_n = 1$ (D) If $l < 1$, then $\lim_{n \rightarrow \infty} a_n = 0$

Question 12 (JAM 2020 • Q12 • ID: JAM_2020_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Define $s_1 = \alpha > 0$ and $s_{n+1} = \sqrt{\frac{1+s_n^2}{1+\alpha}}$, $n \geq 1$. Which of the following is true?

- (A) If $s_n^2 < \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\sqrt{\alpha}}$
- (B) If $s_n^2 < \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\alpha}$
- (C) If $s_n^2 > \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\sqrt{\alpha}}$
- (D) If $s_n^2 > \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\alpha}$

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Question 13 (JAM 2020 • Q13 • ID: JAM_2020_Q13)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 13 Suppose that S is the sum of a convergent series $\sum_{n=1}^{\infty} a_n$. Define $t_n = a_n + a_{n+1} + a_{n+2}$. Then the series $\sum_{n=1}^{\infty} t_n$

- (A) diverges
- (B) converges to $3S - a_1 - a_2$
- (C) converges to $3S - a_1 - 2a_2$
- (D) converges to $3S - 2a_1 - a_2$

Question 14 (JAM 2020 • Q14 • ID: JAM_2020_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 Let $a \in \mathbb{R}$. If $f(x) = \begin{cases} (x+a)^2, & x \leq 0 \\ (x+a)^3, & x > 0, \end{cases}$

then

- (A) $\frac{d^2 f}{dx^2}$ does not exist at $x = 0$ for any value of a
- (B) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly one value of a
- (C) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly two values of a
- (D) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for infinitely many values of a

$$\begin{cases} x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y}, & xy \neq 0 \end{cases}$$

Question 15 (JAM 2020 • Q15 • ID: JAM_2020_Q15)

MCQ

+2 M

Neg: -0.67

Q. 15 Let $f(x, y) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y}, & y \neq 0, x = 0 \\ 0, & x = y = 0. \end{cases}$

Which of the following is true at $(0, 0)$?

- (A) f is not continuous
- (B) $\frac{\partial f}{\partial x}$ is continuous but $\frac{\partial f}{\partial y}$ is not continuous
- (C) f is not differentiable
- (D) f is differentiable but both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not continuous

Question 16 (JAM 2020 • Q16 • ID: JAM_2020_Q16)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 16 Let S be the surface of the portion of the sphere with centre at the origin and radius 4, above the xy -plane. Let $\vec{F} = y\hat{i} - x\hat{j} + yx^3\hat{k}$. If $\hat{n}$ is the unit outward normal to S , then

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

equals

(A) -32π (B) -16π (C) 16π (D) 32π

Question 17 (JAM 2020 • Q17 • ID: JAM_2020_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f(x, y, z) = x^3 + y^3 + z^3 - 3xyz$. A point at which the gradient of the function f is equal to zero is

(A) $(-1, 1, -1)$ (B) $(-1, -1, -1)$ (C) $(-1, 1, 1)$ (D) $(1, -1, 1)$

Question 18 (JAM 2020 • Q18 • ID: JAM_2020_Q18)

MCQ

+2 M

Neg: -0.67

Q. 18 The area bounded by the curves $x^2 + y^2 = 2x$ and $x^2 + y^2 = 4x$, and the straight lines $y = x$ and $y = 0$ is

(A) $3 \left(\frac{\pi}{2} + \frac{1}{4} \right)$ (B) $3 \left(\frac{\pi}{4} + \frac{1}{2} \right)$ (C) $2 \left(\frac{\pi}{4} + \frac{1}{3} \right)$ (D) $2 \left(\frac{\pi}{3} + \frac{1}{4} \right)$

Question 19 (JAM 2020 • Q19 • ID: JAM_2020_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Let M be a real 6×6 matrix. Let 2 and -1 be two eigenvalues of M . If $M^5 = aI + bM$, where $a, b \in \mathbb{R}$, then

(A) $a = 10, b = 11$

(B) $a = -11, b = 10$

(C) $a = -10, b = 11$

(D) $a = 10, b = -11$

Question 20 (JAM 2020 • Q20 • ID: JAM_2020_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Let M be an $n \times n$ ($n \geq 2$) non-zero real matrix with $M^2 = 0$ and let $\alpha \in \mathbb{R} \setminus \{0\}$. Then

(A) α is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$ (B) α is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$ (C) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(M - \alpha I)$ (D) $-\alpha$ is the only eigenvalue of $(M + \alpha I)$ and $(\alpha I - M)$

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Question 21 (JAM 2020 • Q21 • ID: JAM_2020_Q21)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 21 Consider the differential equation $L[y] = (y - y^2)dx + xdy = 0$. The function $f(x, y)$ is said to be an integrating factor of the equation if $f(x, y)L[y] = 0$ becomes exact.

If $f(x, y) = \frac{1}{x^2 y^2}$, then

(A) f is an integrating factor and $y = 1 - kxy$, $k \in \mathbb{R}$ is NOT its general solution(B) f is an integrating factor and $y = -1 + kxy$, $k \in \mathbb{R}$ is its general solution(C) f is an integrating factor and $y = -1 + kxy$, $k \in \mathbb{R}$ is NOT its general solution(D) f is NOT an integrating factor and $y = 1 + kxy$, $k \in \mathbb{R}$ is its general solution

Question 22 (JAM 2020 • Q22 • ID: JAM_2020_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 A solution of the differential equation $2x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} - y = 0, x > 0$ that passes through the point $(1, 1)$ is

(A) $y = \frac{1}{x}$

(B) $y = \frac{1}{x^2}$

(C) $y = \frac{1}{\sqrt{x}}$

(D) $y = \frac{1}{x^{3/2}}$

Question 23 (JAM 2020 • Q23 • ID: JAM_2020_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

(A) If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set(B) If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set(C) If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set(D) If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Question 24 (JAM 2020 • Q24 • ID: JAM_2020_Q24)

MCQ

+2 M

Neg: -0.67

Q. 24 The value of the triple integral $\iiint_V (x^2 y + 1) dx dy dz$, where V is the region given by $x^2 + y^2 \leq 1, 0 \leq z \leq 2$ is

(A) π

(B) 2π

(C) 3π

(D) 4π

Question 25 (JAM 2020 • Q25 • ID: JAM_2020_Q25)

MCQ

+2 M

Neg: -0.67

Q. 25 Let S be the part of the cone $z^2 = x^2 + y^2$ between the planes $z = 0$ and $z = 1$. Then the value of the surface integral $\iint_S (x^2 + y^2) dS$ is

(A) π

(B) $\frac{\pi}{\sqrt{2}}$

(C) $\frac{\pi}{\sqrt{3}}$

(D) $\frac{\pi}{2}$

Question 26 (JAM 2020 • Q26 • ID: JAM_2020_Q26)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 26 Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $x, y, z \in \mathbb{R}$. Which of the following is FALSE?

(A) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$

(B) $\nabla \cdot (\vec{a} \times \vec{r}) = 0$

(C) $\nabla \times (\vec{a} \times \vec{r}) = \vec{a}$

(D) $\nabla \cdot ((\vec{a} \cdot \vec{r})\vec{r}) = 4(\vec{a} \cdot \vec{r})$

Question 27 (JAM 2020 • Q27 • ID: JAM_2020_Q27)

MCQ

+2 M

Neg: -0.67

Q. 27 Let $D = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}$ and $f : D \rightarrow \mathbb{R}$ be a non-constant continuous function. Which of the following is TRUE?

(A) The range of f is unbounded(B) The range of f is a union of open intervals(C) The range of f is a closed interval(D) The range of f is a union of at least two disjoint closed intervals

Question 28 (JAM 2020 • Q28 • ID: JAM_2020_Q28)

MCQ

+2 M

Neg: -0.67

Q. 28 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $f\left(\frac{1}{2}\right) = -\frac{1}{2}$ and

$$|f(x) - f(y) - (x - y)| \leq \sin(|x - y|^2)$$

for all $x, y \in [0, 1]$. Then $\int_0^1 f(x) dx$ is

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

Question 29 (JAM 2020 • Q29 • ID: JAM_2020_Q29)

MCQ

+2 M

Neg: -0.67

Q. 29 Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ be the circle group under multiplication and $i = \sqrt{-1}$. Then the set $\{\theta \in \mathbb{R} : \langle e^{i2\pi\theta} \rangle \text{ is infinite}\}$ is

(A) empty

(B) non-empty and finite

(C) countably infinite

(D) uncountable

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Question 30 (JAM 2020 • Q30 • ID: JAM_2020_Q30)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 30 Let $F = \{\omega \in \mathbb{C} : \omega^{2020} = 1\}$. Consider the groups

$$G = \left\{ \begin{pmatrix} \omega & z \\ 0 & 1 \end{pmatrix} : \omega \in F, z \in \mathbb{C} \right\}$$

and

$$H = \left\{ \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} : z \in \mathbb{C} \right\}$$

under matrix multiplication. Then the number of cosets of H in G is

(A) 1010

(B) 2019

(C) 2020

(D) infinite

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Question 31 (JAM 2020 • Q31 • ID: JAM_2020_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 Let $a, b, c \in \mathbb{R}$ such that $a < b < c$. Which of the following is/are true for any continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(a) = b, f(b) = c$ and $f(c) = a$?

- (A) There exists $\alpha \in (a, c)$ such that $f(\alpha) = \alpha$
- (B) There exists $\beta \in (a, b)$ such that $f(\beta) = \beta$
- (C) There exists $\gamma \in (a, b)$ such that $(f \circ f)(\gamma) = \gamma$
- (D) There exists $\delta \in (a, c)$ such that $(f \circ f \circ f)(\delta) = \delta$

Question 32 (JAM 2020 • Q32 • ID: JAM_2020_Q32)

MSQ

+2 M

Neg: Nil (0)

Q. 32 If $s_n = \frac{(-1)^n}{2^n + 3}$ and $t_n = \frac{(-1)^n}{4n - 1}, n = 0, 1, 2, \dots$, then

- (A) $\sum_{n=0}^{\infty} s_n$ is absolutely convergent
- (B) $\sum_{n=0}^{\infty} t_n$ is absolutely convergent
- (C) $\sum_{n=0}^{\infty} s_n$ is conditionally convergent
- (D) $\sum_{n=0}^{\infty} t_n$ is conditionally convergent

Question 33 (JAM 2020 • Q33 • ID: JAM_2020_Q33)

MSQ

+2 M

Neg: Nil (0)

Q. 33 Let $a, b \in \mathbb{R}$ and $a < b$. Which of the following statement(s) is/are true?

- (A) There exists a continuous function $f : [a, b] \rightarrow (a, b)$ such that f is one-one
- (B) There exists a continuous function $f : [a, b] \rightarrow (a, b)$ such that f is onto
- (C) There exists a continuous function $f : (a, b) \rightarrow [a, b]$ such that f is one-one
- (D) There exists a continuous function $f : (a, b) \rightarrow [a, b]$ such that f is onto

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Question 34 (JAM 2020 • Q34 • ID: JAM_2020_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 34 Let V be a non-zero vector space over a field F . Let $S \subset V$ be a non-empty set. Consider the following properties of S :

- (I) For any vector space W over F , any map $f : S \rightarrow W$ extends to a linear map from V to W .
- (II) For any vector space W over F and any two linear maps $f, g : V \rightarrow W$ satisfying $f(s) = g(s)$ for all $s \in S$, we have $f(v) = g(v)$ for all $v \in V$.
- (III) S is linearly independent.
- (IV) The span of S is V .

Which of the following statement(s) is /are true?

- (A) (I) implies (IV)
- (B) (I) implies (III)
- (C) (II) implies (III)
- (D) (II) implies (IV)

Question 35 (JAM 2020 • Q35 • ID: JAM_2020_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Let $L[y] = x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} + qy$, where p, q are real constants. Let $y_1(x)$ and $y_2(x)$ be two solutions of $L[y] = 0, x > 0$, that satisfy $y_1(x_0) = 1, y_1'(x_0) = 0, y_2(x_0) = 0$ and $y_2'(x_0) = 1$ for some $x_0 > 0$. Then,

- (A) $y_1(x)$ is not a constant multiple of $y_2(x)$
 (B) $y_1(x)$ is a constant multiple of $y_2(x)$
 (C) $1, \ln x$ are solutions of $L[y] = 0$ when $p = 1, q = 0$
 (D) $x, \ln x$ are solutions of $L[y] = 0$ when $p + q \neq 0$

Question 36 (JAM 2020 • Q36 • ID: JAM_2020_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 Consider the following system of linear equations

$$x + y + 5z = 3, \quad x + 2y + mz = 5 \quad \text{and} \quad x + 2y + 4z = k.$$

The system is consistent if

- (A) $m \neq 4$ (B) $k \neq 5$ (C) $m = 4$ (D) $k = 5$

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Question 37 (JAM 2020 • Q37 • ID: JAM_2020_Q37)

MSQ

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 37 Let $a = \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{(n-1)}{n^2} \right)$ and $b = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right)$.

Which of the following is/are true?

- (A) $a > b$ (B) $a < b$ (C) $ab = \ln \sqrt{2}$ (D) $\frac{a}{b} = \ln \sqrt{2}$

Question 38 (JAM 2020 • Q38 • ID: JAM_2020_Q38)

MSQ

+2 M

Neg: Nil (0)

Q. 38 Let S be that part of the surface of the paraboloid $z = 16 - x^2 - y^2$ which is above the plane $z = 0$ and D be its projection on the xy -plane. Then the area of S equals

(A) $\iint_D \sqrt{1 + 4(x^2 + y^2)} \, dx dy$

(B) $\iint_D \sqrt{1 + 2(x^2 + y^2)} \, dx dy$

(C) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} \, r dr d\theta$

(D) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} \, r dr d\theta$

Question 39 (JAM 2020 • Q39 • ID: JAM_2020_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Let f be a real valued function of a real variable, such that $|f^{(n)}(0)| \leq K$ for all $n \in \mathbb{N}$, where $K > 0$. Which of the following is/are true?

(A) $\left| \frac{f^{(n)}(0)}{n!} \right|^{\frac{1}{n}} \rightarrow 0$ as $n \rightarrow \infty$

(B) $\left| \frac{f^{(n)}(0)}{n!} \right|^{\frac{1}{n}} \rightarrow \infty$ as $n \rightarrow \infty$

(C) $f^{(n)}(x)$ exists for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$

(D) The series $\sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{(n-1)!}$ is absolutely convergent

Question 40 (JAM 2020 • Q40 • ID: JAM_2020_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let G be a group with identity e . Let H be an abelian non-trivial proper subgroup of G with the property that $H \cap gHg^{-1} = \{e\}$ for all $g \notin H$.

If $K = \{g \in G : gh = hg \text{ for all } h \in H\}$, then

(A) K is a proper subgroup of H

(B) H is a proper subgroup of K

(C) $K = H$

(D) there exists no abelian subgroup $L \subseteq G$ such that K is a proper subgroup of L

Question 41 (JAM 2020 • Q41 • ID: JAM_2020_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 Let $x_n = n^{\frac{1}{n}}$ and $y_n = e^{1-x_n}$, $n \in \mathbb{N}$. Then the value of $\lim_{n \rightarrow \infty} y_n$ is _____.

Question 42 (JAM 2020 • Q42 • ID: JAM_2020_Q42)

NAT

+1 M

Neg: Nil (0)

Q. 42 Let $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ and S be the sphere given by $(x-2)^2 + (y-2)^2 + (z-2)^2 = 4$. If $\hat{n}$ is the unit outward normal to S , then

$$\frac{1}{\pi} \iint_S \vec{F} \cdot \hat{n} \, dS$$

is _____.

Question 43 (JAM 2020 • Q43 • ID: JAM_2020_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that f, f', f'' are continuous functions with $f > 0$, $f' > 0$ and $f'' > 0$. Then

$$\lim_{x \rightarrow -\infty} \frac{f(x) + f'(x)}{2}$$

is _____.

Question 44 (JAM 2020 • Q44 • ID: JAM_2020_Q44)

NAT

+1 M

Neg: Nil (0)

Q. 44 Let $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ and $f : S \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Then

$$\max \left\{ \delta : \left| x - \frac{1}{3} \right| < \delta \implies \left| f(x) - f\left(\frac{1}{3}\right) \right| < 1 \right\}$$

is _____. (rounded off to two decimal places)

Question 45 (JAM 2020 • Q45 • ID: JAM_2020_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 Let $f(x, y) = e^x \sin y$, $x = t^3 + 1$ and $y = t^4 + t$. Then $\frac{df}{dt}$ at $t = 0$ is _____. (rounded off to two decimal places)

MA

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Question 46 (JAM 2020 • Q46 • ID: JAM_2020_Q46)

NAT

+1 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 46 Consider the differential equation

$$\frac{dy}{dx} + 10y = f(x), \quad x > 0,$$

where $f(x)$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = 1$. Then the value of

$$\lim_{x \rightarrow \infty} y(x)$$

is _____.

Question 47 (JAM 2020 • Q47 • ID: JAM_2020_Q47)

NAT

+1 M

Neg: Nil (0)

Q. 47 If $\int_0^1 \int_{2y}^2 e^{x^2} dx dy = k(e^4 - 1)$, then k equals _____.

Question 48 (JAM 2020 • Q48 • ID: JAM_2020_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $f(x, y) = 0$ be a solution of the homogeneous differential equation

$$(2x + 5y)dx - (x + 3y)dy = 0.$$

If $f(x + \alpha, y - 3) = 0$ is a solution of the differential equation

$$(2x + 5y - 1)dx + (2 - x - 3y)dy = 0,$$

then the value of α is _____.

Question 49 (JAM 2020 • Q49 • ID: JAM_2020_Q49)

NAT

+1 M

Neg: Nil (0)

Q. 49 Consider the real vector space $P_{2020} = \{\sum_{i=0}^n a_i x^i : a_i \in \mathbb{R} \text{ and } 0 \leq n \leq 2020\}$. Let W be the subspace given by

$$W = \left\{ \sum_{i=0}^n a_i x^i \in P_{2020} : a_i = 0 \text{ for all odd } i \right\}.$$

Then, the dimension of W is _____.

Question 50 (JAM 2020 • Q50 • ID: JAM_2020_Q50)

NAT

+1 M

Neg: Nil (0)

Q. 50 Let $\phi : S_3 \rightarrow S^1$ be a non-trivial non-injective group homomorphism. Then, the number of elements in the kernel of ϕ is _____.

MA

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Question 51 (JAM 2020 • Q51 • ID: JAM_2020_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 The sum of the series $\frac{1}{2(2^2 - 1)} + \frac{1}{3(3^2 - 1)} + \frac{1}{4(4^2 - 1)} + \dots$ is _____.

Question 52 (JAM 2020 • Q52 • ID: JAM_2020_Q52)

NAT

+2 M

Neg: Nil (0)

Q. 52 Consider the expansion of the function $f(x) = \frac{3}{(1-x)(1+2x)}$ in powers of x , that is valid in $|x| < \frac{1}{2}$. Then the coefficient of x^4 is _____.

Question 53 (JAM 2020 • Q53 • ID: JAM_2020_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 The minimum value of the function $f(x, y) = x^2 + xy + y^2 - 3x - 6y + 11$ is _____.

Question 54 (JAM 2020 • Q54 • ID: JAM_2020_Q54)

NAT

+2 M

Neg: Nil (0)

Q. 54 Let $f(x) = \sqrt{x} + \alpha x$, $x > 0$ and

$$g(x) = a_0 + a_1(x-1) + a_2(x-1)^2$$

be the sum of the first three terms of the Taylor series of $f(x)$ around $x = 1$. If $g(3) = 3$, then α is _____.

Question 55 (JAM 2020 • Q55 • ID: JAM_2020_Q55)

NAT

+2 M

Neg: Nil (0)

Q. 55 Let C be the boundary of the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$ oriented in the counter clockwise sense. Then, the value of the line integral

$$\oint_C x^2 y^2 dx + (x^2 - y^2) dy$$

is _____. (rounded off to two decimal places)

Question 56 (JAM 2020 • Q56 • ID: JAM_2020_Q56)

NAT

+2 M

Neg: Nil (0)

Q. 56 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f'(x) = f(x)$ for all x . Suppose that $f(\alpha x)$ and $f(\beta x)$ are two non-zero solutions of the differential equation

$$4 \frac{d^2 y}{dx^2} - p \frac{dy}{dx} + 3y = 0$$

satisfying

$$f(\alpha x)f(\beta x) = f(2x) \text{ and } f(\alpha x)f(-\beta x) = f(x).$$

Then, the value of p is _____.

MA

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Question 57 (JAM 2020 • Q57 • ID: JAM_2020_Q57)

NAT

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 57 If $x^2 + xy^2 = c$, where $c \in \mathbb{R}$, is the general solution of the exact differential equation

$$M(x, y) dx + 2xy dy = 0,$$

then $M(1, 1)$ is _____.

$$\begin{bmatrix} 9 & 2 & 7 & 1 \end{bmatrix}$$

Question 58 (JAM 2020 • Q58 • ID: JAM_2020_Q58)

NAT

+2 M

Neg: Nil (0)

Q. 58 Let $M = \begin{bmatrix} 9 & 2 & 7 & 1 \\ 0 & 7 & 2 & 1 \\ 0 & 0 & 11 & 6 \\ 0 & 0 & -5 & 0 \end{bmatrix}$. Then, the value of $\det((8I - M)^3)$ is _____.

Question 59 (JAM 2020 • Q59 • ID: JAM_2020_Q59)

NAT

+2 M

Neg: Nil (0)

Q. 59 Let $T : \mathbb{R}^7 \rightarrow \mathbb{R}^7$ be a linear transformation with $\text{Nullity}(T) = 2$. Then, the minimum possible value for $\text{Rank}(T^2)$ is _____.

Question 60 (JAM 2020 • Q60 • ID: JAM_2020_Q60)

NAT

+2 M

Neg: Nil (0)

Q. 60 Suppose that G is a group of order 57 which is NOT cyclic. If G contains a unique subgroup H of order 19, then for any $g \notin H$, $o(g)$ is _____.

END OF THE QUESTION PAPER

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2020 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	C
2	MCQ	+1	D
3	MCQ	+1	D
4	MCQ	+1	C
5	MCQ	+1	Marks to All
6	MCQ	+1	C
7	MCQ	+1	D
8	MCQ	+1	D
9	MCQ	+1	D
10	MCQ	+1	B
11	MCQ	+2	D
12	MCQ	+2	A
13	MCQ	+2	D
14	MCQ	+2	A
15	MCQ	+2	D
16	MCQ	+2	A
17	MCQ	+2	B
18	MCQ	+2	B
19	MCQ	+2	A
20	MCQ	+2	B

Q. No.	Type	Marks	Official Key
21	MCQ	+2	C
22	MCQ	+2	A
23	MCQ	+2	D
24	MCQ	+2	B
25	MCQ	+2	B
26	MCQ	+2	C
27	MCQ	+2	C
28	MCQ	+2	A
29	MCQ	+2	D
30	MCQ	+2	C
31	MSQ	+2	A;C;D
32	MSQ	+2	A;D
33	MSQ	+2	A;C;D
34	MSQ	+2	B;D
35	MSQ	+2	A;C
36	MSQ	+2	A;D
37	MSQ	+2	B;C
38	MSQ	+2	A;D
39	MSQ	+2	A;D
40	MSQ	+2	C;D

Q. No.	Type	Marks	Official Key
41	NAT	+1	1 to 1
42	NAT	+1	32 to 32
43	NAT	+1	0 to INF
44	NAT	+1	0.08 to 0.09
45	NAT	+1	2.70 to 2.72
46	NAT	+1	0.1 to 0.1
47	NAT	+1	0.25 to 0.25
48	NAT	+1	7 to 7
49	NAT	+1	1011 to 1011
50	NAT	+1	3 to 3
51	NAT	+2	0.25 to 0.25
52	NAT	+2	33 to 33
53	NAT	+2	2 to 2
54	NAT	+2	0.5 to 0.5
55	NAT	+2	0.65 to 0.67
56	NAT	+2	8 to 8
57	NAT	+2	3 to 3
58	NAT	+2	-216 to -216
59	NAT	+2	3 to 3
60	NAT	+2	3 to 3

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, -2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.